

STAT 4530/6530: Homework 4

Aniket Ghosh
STAT 6530

March 27

1. (from Gelman et al., pg. 37) Placenta previa is an unusual pregnancy condition in which the placenta is implanted very low in the uterus, obstructing the fetus from a normal vaginal delivery. A study of the sex of placenta previa births in Germany found that there were 437 females among 980 placenta previa births. Let θ be the true proportion of female births in the placenta previa population.

(a) (2 points) What sampling model most appropriately describes Y , the number of females among the 980 placenta previa births from the Germany study?

Solution:

The most appropriate sampling model for Y , the number of females among the 980 placenta previa births, is a binomial distribution, expressed as:

$$Y|\theta \sim \text{Binomial}(n = 980, \theta)$$

where n is the total number of births and θ is the true proportion of female births in the placenta previa population.

(b) (4 points) Researcher A assumes there is no prior information about θ . Specify the appropriate prior distribution for θ and identify the posterior distribution (using derived results from class). Plot the prior and posterior distribution together in R.

Solution:

If Researcher A assumes there is no prior information, we should take a non-informative Uniform prior over the interval $[0, 1]$. This is equivalent to a Beta distribution with $\alpha = 1$ and $\beta = 1$:

$$\theta \sim \text{Beta}(1, 1)$$

Using the derived results for the Beta-Binomial conjugacy, the posterior distribution is $\text{Beta}(\alpha + y, \beta + n - y)$. Given $y = 437$ and $n = 980$:

$$\theta|y \sim \text{Beta}(1 + 437, 1 + 980 - 437) \implies \text{Beta}(438, 544)$$

Plotting the Prior and Posteriors:

```
1 # Researcher A: Prior and Posterior
2 theta <- seq(0, 1, length.out = 1000)
3 priorA <- dbeta(theta, 1, 1)
4 postA <- dbeta(theta, 438, 544)
```

5

```

6 plot(theta, postA, type = "l", col = "blue", lwd = 2,
7       ylab = "Density", xlab = expression(theta),
8       main = "Researcher A: Prior vs Posterior")
9 lines(theta, priorA, col = "red", lwd = 2, lty = 2)
10 legend("topright", legend = c("Prior Beta(1,1)", "Posterior Beta(438,544)"),
11        col = c("red", "blue"), lty = c(2, 1), lwd = 2)

```

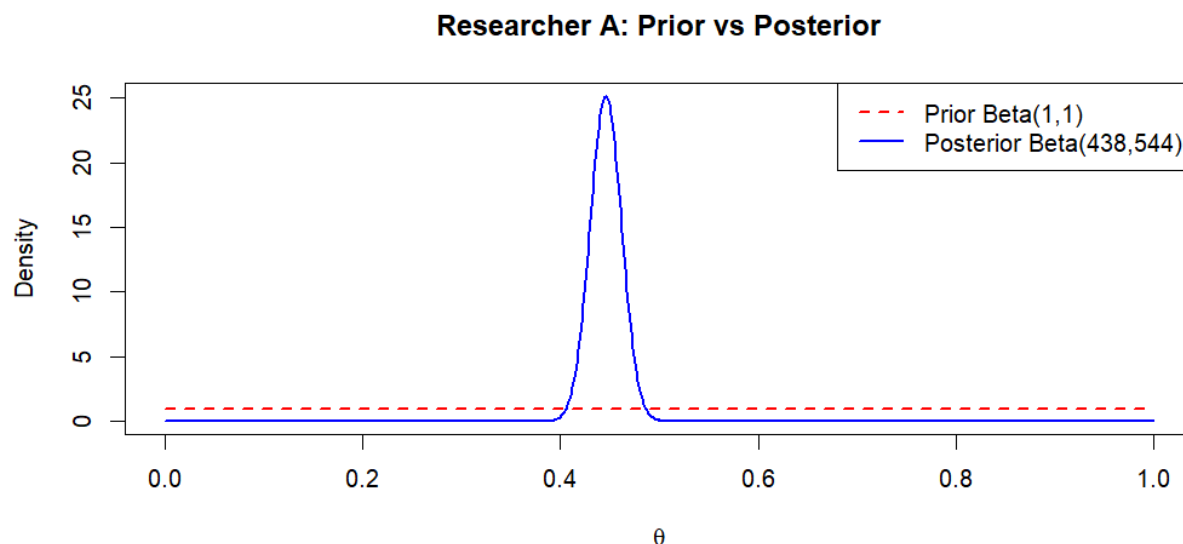


Figure 1. Researcher A: Prior vs Posterior

(c) (4 points) Researcher B worked on a confidential pilot study, which saw 120 females out of 200 placenta previa births in Austria. Specify the appropriate prior distribution for θ and identify the posterior distribution (using derived results from class). Plot the prior and posterior distribution together in R.

Solution:

Researcher B has prior information from a pilot study showing 120 females out of 200 births. We can write this directly as a Beta prior where $\alpha = 120$ and $\beta = 200 - 120 = 80$:

$$\theta \sim \text{Beta}(120, 80)$$

Updating this prior with the new data ($y = 437$, $n = 980$) to find the posterior:

$$\theta|y \sim \text{Beta}(120 + 437, 80 + 980 - 437) \implies \text{Beta}(557, 623)$$

```

1 # Researcher B: Prior and Posterior
2 priorB <- dbeta(theta, 120, 80)
3 postB <- dbeta(theta, 557, 623)
4
5 plot(theta, postB, type = "l", col = "blue", lwd = 2,
6       ylab = "Density", xlab = expression(theta),
7       main = "Researcher B: Prior vs Posterior")
8 lines(theta, priorB, col = "red", lwd = 2, lty = 2)
9 legend("topright", legend = c("Prior Beta(120,80)", "Posterior Beta(557,623)"),
10        col = c("red", "blue"), lty = c(2, 1), lwd = 2)

```

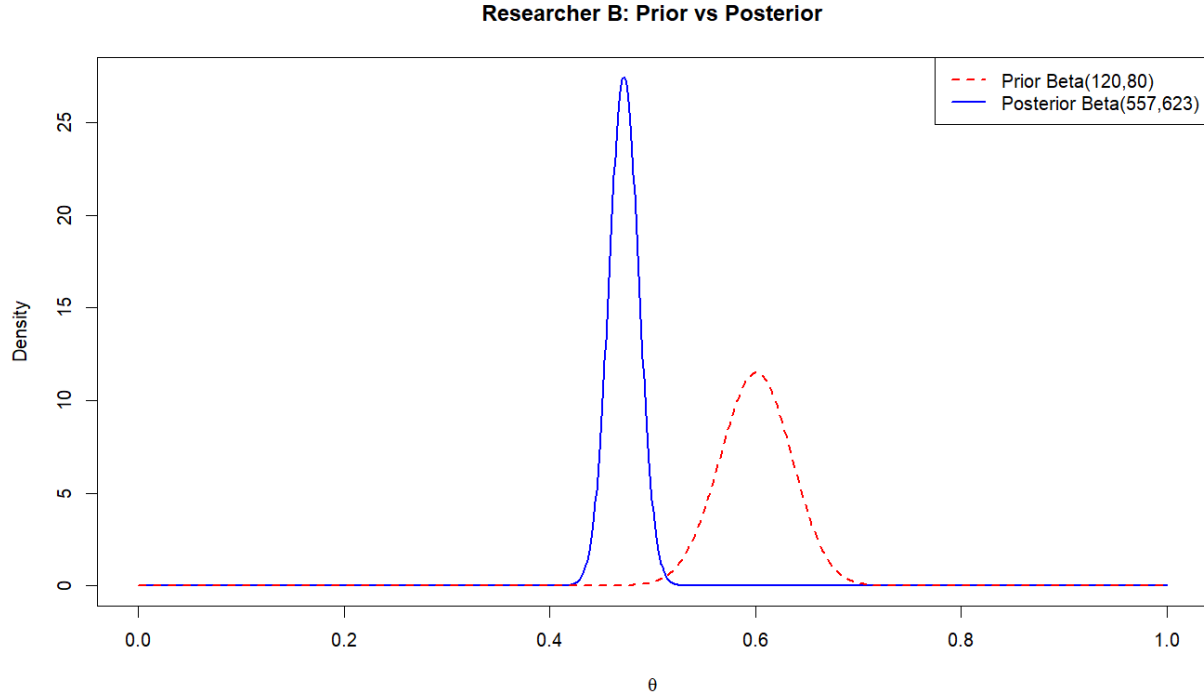


Figure 2. Researcher B: Prior vs Posterior

(d) (4 points) For both Researchers A and B, use known formulas to find the posterior mean and variance. In each case, identify the posterior mean weights attributed to the prior and the observed data.

Solution:

Both have Beta(α, β) posteriors, for whom the mean is $\frac{\alpha}{\alpha+\beta}$ and the variance is $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

- **Researcher A:**

$$\text{Mean} = \frac{438}{438 + 544} \approx 0.446$$

$$\text{Variance} = \frac{438 \times 544}{(982)^2(983)} \approx 0.00025$$

$$\text{Weights: Prior weight} = \frac{\alpha+\beta}{\alpha+\beta+n} = \frac{2}{982} \approx 0.002. \text{ Data weight} = \frac{n}{\alpha+\beta+n} = \frac{980}{982} \approx 0.998.$$

- **Researcher B:**

$$\text{Mean} = \frac{557}{557 + 623} \approx 0.472$$

$$\text{Variance} = \frac{557 \times 623}{(1180)^2(1181)} \approx 0.00021$$

$$\text{Weights: Prior weight} = \frac{200}{1180} \approx 0.169. \text{ Data weight} = \frac{980}{1180} \approx 0.831.$$

(e) (4 points) For both Researchers A and B, find a 95% percentile-based posterior credible interval for θ (in R).

Solution:

```
1 # 95% Credible Intervals
2 ci_A <- qbeta(c(0.025, 0.975), 438, 544)
3 ci_B <- qbeta(c(0.025, 0.975), 557, 623)
4 ci_A
5 ci_B
```

Confidence Interval for A : (0.4150655, 0.4771998)

Confidence Interval for B : (0.4436122, 0.5005453)

(f) (4 points) How much evidence does this data provide for the claim that the proportion of female births in the placenta previa population is less than the proportion of female births in the general population, which is approximately 0.485? Answer this question by computing posterior probabilities (in R) for $\theta < 0.485$, using the posteriors for both Researchers A and B.

Solution:

We want to find $P(\theta < 0.485|y)$.

```
1 # Probability theta < 0.485
2 prob_A <- pbeta(0.485, 438, 544)
3 prob_B <- pbeta(0.485, 557, 623)
4 prob_A
5 prob_B
```

Researcher A's model provides very strong evidence (0.992826%) that the proportion is less than 0.485. On the other hand, Researcher B's model provides moderate evidence for the same (0.8139057%).

(g) (4 points) Using all of the previous parts, summarize the above Bayesian analyses performed by Researchers A and B, being sure to comment on the influence of prior information on the posterior.

Solution:

We see that Researcher A used an uninformative prior, hence their posterior is almost entirely driven by the German data ($n = 980$), leading to a lower mean estimate (~ 0.446) and high certainty that $\theta < 0.485$. Researcher B, on the other hand, used a much more informative prior from a previous study. Because the prior mean ($120/200 = 0.60$) was significantly higher than the data mean ($437/980 \approx 0.446$), Researcher B's posterior mean was pulled towards the prior mean, up to 0.472. This shows that the strength of prior information can heavily influence the posterior.

(h) (4 points) What might potentially make you weary about the prior used by Researcher B? Explain how you could adjust the prior in light of this.

Solution:

Researcher B's prior comes from a pilot study which was conducted in a different country, i.e. Austria, when we are finding the posterior for Germany. Although they might be similar countries, it's very likely that there are factors involved that do vary, and have not been accounted for in this simple posterior calculation. To adjust for this, Researcher B could down-weight the prior by keeping the same mean but using a smaller effective sample size like $\alpha = 12, \beta = 8$.

2. Thus far, if we have binomial data, then we have explored using the prior distribution $\theta \sim \text{Beta}(a, b)$. The probability density function of this distribution is given by $p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}$ for $0 < \theta < 1$.

(a) (4 points) Show that a $\text{Beta}(1, 1)$ distribution is equivalent to a $\text{Unif}[0,1]$ distribution, which has probability density function $p(\theta) = 1$ for $0 < \theta < 1$.

Solution:

The Beta distribution PDF is $p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}$. Substituting $a = 1$ and $b = 1$, and knowing that $\Gamma(1) = 0! = 1$ and $\Gamma(2) = 1! = 1$:

$$p(\theta) = \frac{\Gamma(2)}{\Gamma(1)\Gamma(1)}\theta^{1-1}(1-\theta)^{1-1} = \frac{1}{1 \cdot 1}\theta^0(1-\theta)^0 = 1$$

This gives us the PDF of a Uniform[0,1] distribution, $p(\theta) = 1$ for $0 < \theta < 1$.

(b) (4 points) Prove that for $\theta \sim \text{Beta}(a, b)$ the expected value is: $E(\theta) = \frac{a}{a+b}$.

Solution:

By definition, the expected value is:

$$E(\theta) = \int_0^1 \theta \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}d\theta$$

$$E(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \theta^a(1-\theta)^{b-1}d\theta$$

We can consider the integral as an unnormalized Beta function for a $\text{Beta}(a+1, b)$ distribution. Hence, the integral evaluates to $\frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}$. Therefore:

$$E(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \cdot \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}$$

$$E(\theta) = \frac{(a+b-1)!}{(a-1)!(b-1)!} \cdot \frac{a!(b-1)!}{(a+b)!} = \frac{a}{a+b}$$

(c) (4 points) In R, plot the beta distribution when $a = b = 1$, $a = b = 5$, and $a = b = 10$. Which of these hyperparameter settings would serve to provide the most prior information?

Solution:

```
1 theta <- seq(0, 1, length.out=100)
2 plot(theta, dbeta(theta, 1, 1), type="l", col="black", ylim=c(0,4), ylab="Density"
3 )
4 lines(theta, dbeta(theta, 5, 5), col="blue")
5 lines(theta, dbeta(theta, 10, 10), col="red")
6 legend("topright", legend=c("a=b=1", "a=b=5", "a=b=10"),
7       col=c("black", "blue", "red"), lty=1)
```

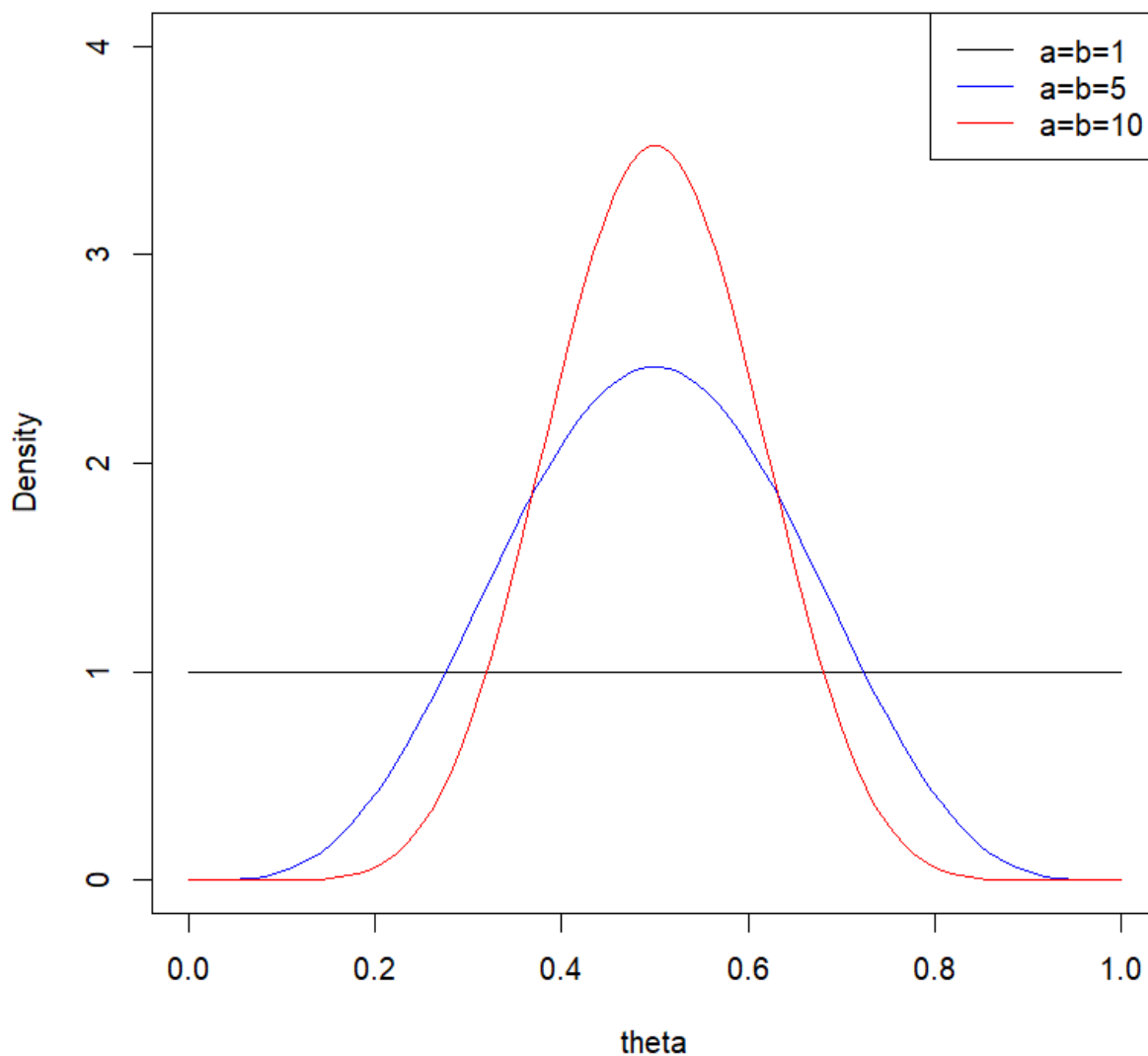


Figure 3. Plotting Beta Distributions for varying Hyperparameters

The setting $a = b = 10$ provides the most prior information, as it has the lowest variance and is highly concentrated around the mean of 0.5.

3. Suppose a licorice candy manufacturer produces pieces which are either red or black. Let θ be the true proportion of pieces which are red. You are tasked with estimating θ .

(a) (4 points) You have done some investigative research, and have found that many believe this candy manufacturer packages licorice pieces in two different ways: individual packages are aimed to contain either 30% red, 70% black pieces, or 70% red,

30% black pieces. Describe what a prior distribution might look like which adequately describes this prior information. Why would the conjugate beta prior be inappropriate here?

Solution:

The prior distribution in this case is bimodal, with two distinct peaks at $\theta = 0.30$ and $\theta = 0.70$. A conjugate Beta prior is inappropriate here because a Beta distribution is unimodal. It cannot reflect a distribution with two distinct peaks in the interior of the interval.

(b) (4 points) It is challenging to encode the prior information from part (a) into a closed-form mathematical expression or distribution. However, we can still specify prior beliefs numerically. Plot this prior distribution. What value would you change in the code above if you wanted to increase the "informative-ness" of the prior? In your answer for this question, please attach a saved version of your plot, appropriately labeled.

Solution:

```
1 grid <- seq(0,1, length.out=1000)
2 prior_pdf <- 0.5*dnorm(grid, mean=0.3, sd=0.05) + 0.5*dnorm(grid, mean=0.7, sd
  =0.05)
3 prior_pdf <- prior_pdf / sum(prior_pdf)
4
5 plot(grid, prior_pdf, type="l", main="Bimodal Prior Distribution",
6       xlab=expression(theta), ylab="Density")
```

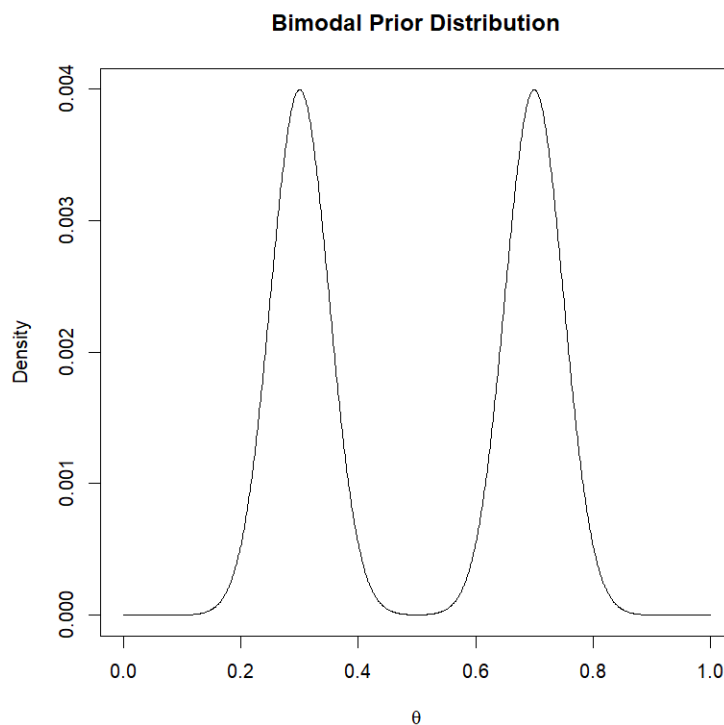


Figure 4. Bimodal Prior Distribution

To increase the "informative-ness" of the prior, I would decrease the standard deviation in the code. For example, changing `sd=0.05` to `sd=0.01` would make the peaks much narrower and sharper, reflecting higher certainty in those specific values.

```
1 grid <- seq(0,1, length.out=1000)
2 prior_pdf <- 0.5*dnorm(grid, mean=0.3, sd=0.01) + 0.5*dnorm(grid, mean=0.7, sd
  =0.01)
3 prior_pdf <- prior_pdf / sum(prior_pdf)
4
5 plot(grid, prior_pdf, type="l", main="Bimodal Prior Distribution",
6       xlab=expression(theta), ylab="Density")
```

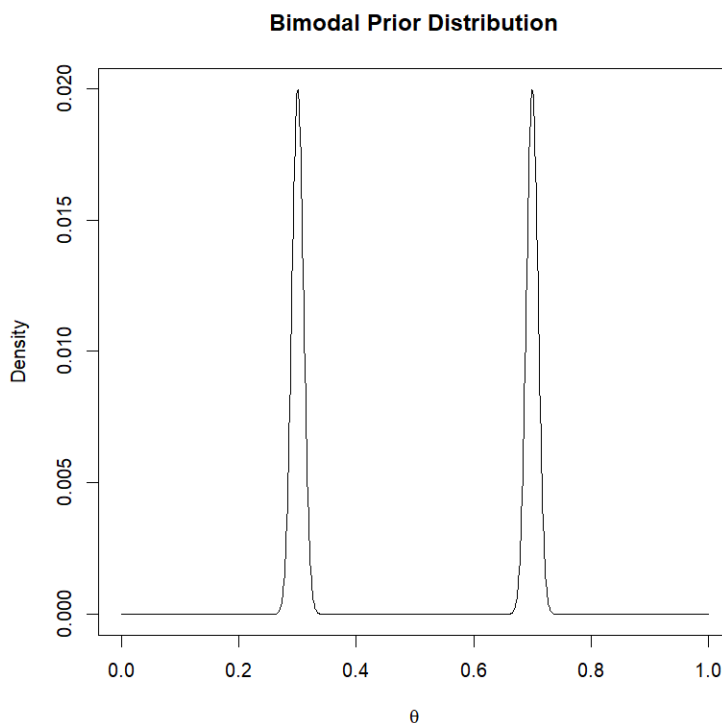


Figure 5. Bimodal Prior Distribution with higher Certainty

(c) (6 points) Let Y be the number of red licorice pieces out of a sample of size n . Assuming a binomial sampling model with the prior specified in part (b), use Bayes' Theorem to write the posterior density $p(\theta|Y)$, up to a proportionality constant. Is this a recognizable closed-form distribution?

Solution:

According to Bayes' Theorem, $p(\theta|y) \propto p(y|\theta)p(\theta)$.

$$p(\theta|y) \propto [\theta^y(1-\theta)^{n-y}] \times \left[\frac{1}{2}e^{-200(\theta-0.3)^2} + \frac{1}{2}e^{-200(\theta-0.7)^2} \right]$$

As we can see, this is not a recognizable closed-form distribution. It is a complex distribution.

4. This question is only required for students enrolled in STAT 6530. (10 points)
Even though we do not get a recognizable closed-form distribution for the posterior in Question 3, part (c), we can still approximate and visualize the posterior using a technique called grid approximation. ... Consider the following three sets of observed data:

- (a) Sample of 25 pieces, 13 of which are red
- (b) Sample of 30 pieces, 15 of which are red
- (c) Sample of 300 pieces, 150 of which are red

For each setting, use the above code to perform grid approximation of the posterior for θ (you will need to specify y and n before running the code). Plot each posterior, and summarize your observations in each scenario.

```

1 # Function to run grid approximation and plot
2 grid_approx <- function(y, n, title) {
3   grid <- seq(0, 1, length.out=1000)
4   prior_pdf <- 0.5*dnorm(grid, 0.3, 0.05) + 0.5*dnorm(grid, 0.7, 0.05)
5   bin_lik <- dbinom(y, n, grid)
6   post_pdf <- bin_lik * prior_pdf
7   post_pdf <- post_pdf / sum(post_pdf)
8
9   plot(grid, post_pdf, type="l", main=title, xlab=expression(theta), ylab="Density
10  ")
11 }
12 # (a) y = 13, n = 25
13 grid_approx(13, 25, "Posterior: 13/25 Red")
14
15 # (b) y = 15, n = 30
16 grid_approx(15, 30, "Posterior: 15/30 Red")
17
18 # (c) y = 150, n = 300
19 grid_approx(150, 300, "Posterior: 150/300 Red")

```

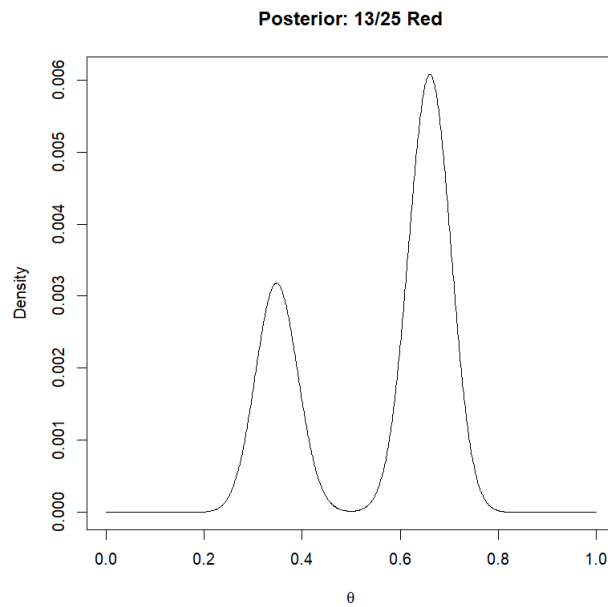


Figure 6. Posterior for 13/25 Red

(a) 13/25 Red: The sample proportion is ~ 0.52 . This pulls the mass toward the center, but our bimodal prior forces the posterior to remain bimodal, split between the 0.3 and 0.7 peaks, as $n = 25$ is not large enough to completely overwhelm our strong prior.

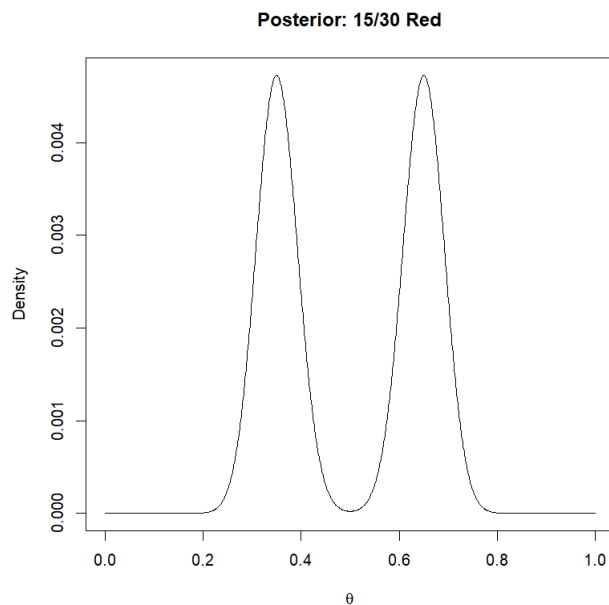


Figure 7. Posterior for 15/30 Red

(b) 15/30 Red: The sample proportion is exactly 0.50. The posterior is symmetric but still bimodal, much like in part a.

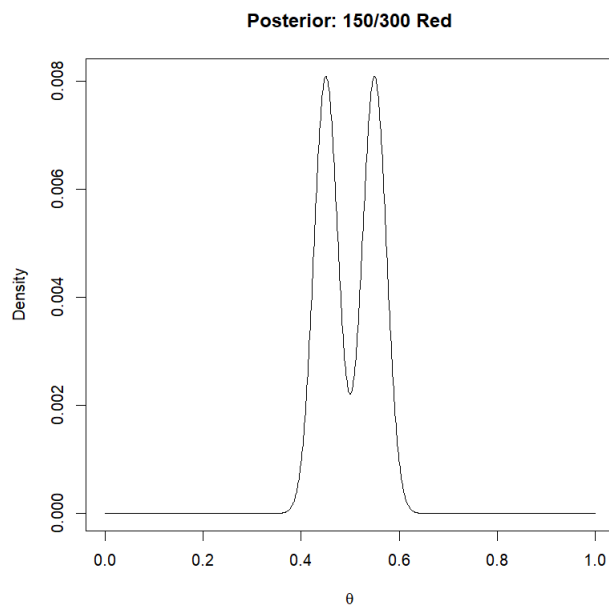


Figure 8. Posterior for 50/300 Red

(c) 150/300 Red: The sample proportion is exactly 0.50, but with $n = 300$, hence the likelihood is massively stronger. The data overwhelms the bimodal prior, forcing our posterior to bring both nodes very close, concentrated around 0.50, but still retaining two different peaks.